

Technical Report

**Edge-Based Anomaly Detection For Fleet
Based Condition Monitoring of Machines**

Submitted By

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List of Abbreviations

- MQTT** Message Queuing Telemetry Transport
- M2M** Machine to Machine
- UI** User Interface

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1 Abstract

Industrial machinery demands maximum reliability, as unexpected breakdowns can lead to production losses, high maintenance costs, and safety hazards. Traditional condition monitoring systems rely heavily on handcrafted rules or supervised models, both of which require extensive expert involvement and large labeled datasets. These limitations make them unsuitable for large, diverse machine fleets. To address this, the proposed work develops an edge-based anomaly detection framework integrated within an edge–fog–cloud architecture for real-time condition monitoring. The system performs local preprocessing, clustering, and unsupervised anomaly detection directly at the edge, reducing latency and bandwidth use while preserving data privacy. The fog layer monitors edge-node health using MQTT-based heartbeat signals and compiles fleet-wide status reports for the cloud. The cloud layer hosts a responsive dashboard for real-time fleet visualization and anomaly tracking. An unsupervised learning approach using hierarchical clustering identifies anomalies without requiring labeled failure data, assigning scores based on inter-machine similarity. Experimental simulations demonstrate that this distributed framework achieves low-latency fault detection, scalable fleet management, and interpretable anomaly visualization, making it well-suited for industrial IoT environments with limited connectivity.

Keywords: Anomaly Detection, Agglomerative Clustering, Fleet Monitoring, Vibration Sensing, Edge

2 Introduction

2.1 Background

Machine fault detection is critical importance in industrial environments, machines are expected to work with maximum reliability — there is no room for failure. When breakdowns occur, the consequences go far beyond technical faults:

- Production losses and unexpected downtime
- High repair and maintenance costs
- Serious injuries or loss of life
- Environmental damage

2.2 Drawbacks of Existing Solutions

Handcrafting Indicators

- Time-consuming – Setting manual vibration thresholds for each motor type requires weeks of expert tuning.
- Not scalable – Each new machine or setup needs its own set of custom rules, in addition to its working conditions (location, humidity, temperature).

Supervised Learning

- Requires large historical labeled datasets – A model needs thousands of examples labeled faulty vs normal to train and validate properly.
- Data diversity – Data is needed for every fault type
- Still needs human involvement – Experts must annotate historical logs, validate model predictions, and adjust settings regularly.

Semi-Supervised Learning

- Ignores environmental factors – A model might misclassify elevated temperatures due to seasonal heat as a machine fault.
- Limited real-time adaptability – Adapting to a sudden change in workload or operational pattern requires retraining, not just reconfiguration.

2.3 Motivation

Despite significant advancements in condition monitoring and predictive maintenance, most existing machine fault detection systems are heavily based on cloud-based analysis or supervised learning models. These approaches face several limitations, such as the need for large labeled datasets, high communication bandwidth, and increased latency due to centralized processing. In real industrial environments where machines operate continuously and often in remote locations, these constraints make cloud-only approaches impractical.

Furthermore, traditional models often fail to consider the varying environmental and operational contexts in which machines operate. For example, a temperature anomaly might simply be the result of seasonal variation rather than an actual fault. Such scenarios highlight the necessity of local decision-making closer to the data source.

Edge computing offers a promising solution that enables computation and analysis at or near the data origin. By processing sensor data locally at the edge node, faults can be detected in real time without depending on constant cloud connectivity. This not only reduces latency and bandwidth usage but also enhances data privacy and system resilience.

2.4 Problem Definition

There is a need for a distributed and adaptive approach that enables local processing, minimizes dependency on the cloud, and performs unsupervised fault detection across fleets of machines. The goal of this work is to design an edge-fog-cloud architecture that supports real-time anomaly detection, reduces communication overhead, and enhances reliability in industrial condition monitoring systems.

3 Literature Survey

3.1 Fleet-Based Anomaly Detection Frameworks

The foundation of fleet-based condition monitoring lies in leveraging similarities across machines to improve anomaly detection without requiring extensive labeled historical data. Hendrickx et al. [1, 2] present a pioneering general anomaly detection framework specifically designed for fleet-based condition monitoring of machines. Their approach utilizes unsupervised learning methodologies that capitalize on the commonalities between similar machines within a fleet, enabling effective anomaly detection without reliance on historical labeled training data. This framework provides real-time online monitoring with integrated visualization capabilities, addressing a critical gap in industrial applications where labeled datasets are prohibitively expensive to acquire. Dhada et al. [3] advance this concept further by proposing hierarchical models for anomaly detection in fleets of industrial assets. Their work demonstrates how hierarchical structures can effectively capture relationships between different equipment types and their collective operational patterns, enabling more sophisticated fault detection mechanisms. Together, these studies establish that fleet-based approaches substantially outperform traditional single-machine models by exploiting the inherent correlations present across multiple similar devices operating under comparable conditions.

3.2 Unsupervised and Real-Time Anomaly Detection Methodologies

Beyond fleet-level frameworks, specialized methodologies have emerged for unsupervised anomaly detection tailored to specific industrial domains. Carratù et al. [5] introduce a novel unsupervised methodology for detecting anomalies in industrial machinery by analyzing electrical current values, a readily available sensor modality in most operational environments. Their approach eliminates the dependency on labeled training data while providing actionable insights into machinery health. Derse et al. [6] contribute domain-specific knowledge through their anomaly detection study on automotive sensor data, demonstrating how multi-modal sensor fusion can effectively identify abnormal vehicle system behaviors. Gou et al. [4] bridge theoretical anomaly detection with practical industry implementation by developing statistical approaches to establish engine system OBD (On-Board Diagnostic) thresholds. Their work demonstrates the importance of combining machine learning anomaly detection with domain expertise to define operationally meaningful thresholds that enable timely maintenance in-

terventions. These methodologies collectively highlight the necessity of domain-specific customization while maintaining the principle of minimal labeled data dependency.

3.3 Federated Learning and Edge-Based Anomaly Detection

As industrial IoT networks scale to encompass thousands of distributed sensors and devices, privacy-preserving distributed learning approaches have gained prominence. Truong et al. [7] propose a lightweight federated learning framework specifically designed for time-series anomaly detection in industrial control systems. Their work addresses critical concerns regarding data privacy and network efficiency by enabling model training at the edge without centralizing sensitive operational data. The International Journal of Computational and Experimental Science and Engineering [8] presents comprehensive research on anomaly detection in IoT networks using federated machine learning approaches, demonstrating the viability of distributed learning paradigms for large-scale industrial deployments. Comprehensive surveys by Pang et al. [9] and Huang et al. [10] provide extensive reviews of deep learning techniques for anomaly detection, covering various architectures and methodologies. Pang et al. [9] examine the landscape of deep learning approaches for anomaly detection across diverse applications, while Huang et al. [10] present recent advancements and emerging trends in deep learning-based anomaly detection. These surveys collectively demonstrate that deep learning combined with federated approaches represents the current frontier in scalable, privacy-preserving anomaly detection for distributed industrial systems.

3.4 Challenges

Condition monitoring of machine fleets presents several critical challenges that necessitate novel approaches. First, the acquisition of labeled historical data remains prohibitively expensive and time-consuming in industrial environments, as domain experts must manually annotate anomalous events in operational data [1, 2]. Second, different machines within a fleet exhibit varying operational characteristics and degradation patterns, requiring frameworks that can adapt across heterogeneous equipment without retraining on each individual unit [3]. Third, real-time monitoring demands low-latency anomaly detection with immediate visualization capabilities to enable rapid intervention before catastrophic failures occur [1, 2].

Additionally, the scalability of anomaly detection systems becomes problematic when deployed across large industrial fleets with thousands of sensors gen-

erating continuous data streams [7, 8]. Privacy and security concerns emerge when deploying centralized learning approaches, as sensitive operational data from multiple facilities may need to be transmitted to cloud infrastructure [7, 8]. Furthermore, the non-stationary nature of industrial sensor data, where operating conditions and equipment behavior change over time, poses difficulties for models trained on historical baselines [9, 10]. Finally, domain-specific challenges such as identifying meaningful anomalies in automotive sensor data and establishing appropriate thresholds for engine diagnostics require expertise in both machine learning and equipment engineering [4, 5, 6].

3.5 Research Gap

Table 1: Research Gaps in Fleet-Based Anomaly Detection and Condition Monitoring

Research Gap	Description	References
Fleet-based Unsupervised Anomaly Detection	Most existing approaches either rely on supervised learning with labeled datasets or require extensive historical baselines. Limited frameworks leverage cross-machine similarity without requiring historical labeled data.	[1, 2, 3, 9, 10]
Federated Learning Integration	While federated learning has emerged as a privacy-preserving approach, its integration with real-time anomaly detection in industrial IoT networks remains nascent. Few studies address the combination with resource-constrained edge devices.	[7, 8]
Generalized Electrical Current Anomaly Detection	Development of generalizable electrical current-based anomaly detection methodologies for diverse industrial machinery types is limited. Most work focuses on specific equipment categories.	[5, 6]
Automotive Sensor Thresholds	Gap in practical methodologies for automotive sensor systems bridging theoretical machine learning advancements with industry-applicable operational thresholds.	[4, 6]
Real-time Fleet-based Monitoring	Insufficient focus on domain-specific applications to fleet-based condition monitoring systems that operate without historical labeled data and provide real-time visualization.	[1, 2, 9, 10]

This research gap underscores the need for an integrated framework that combines fleet-level learning, unsupervised techniques, real-time processing, and practical domain applicability.

4 Problem Formulation

4.1 System Overview

In industrial environments, unexpected machine failures can result in significant downtime, costly repairs, and even safety hazards. Traditional condition monitoring methods depend heavily on handcrafted fault indicators and extensive labeled historical datasets. However, obtaining such data is often impractical because machines operate under numerous varying conditions, and potential faults are rare or unknown beforehand.

Furthermore, most existing artificial intelligence-based monitoring systems analyze only single machines and function as black-box models, offering limited interpretability to domain experts.

The problem, therefore, is to develop a general and interpretable anomaly detection framework capable of identifying faulty machines in a fleet without requiring labeled or historical data. The framework must be able to:

1. Continuously monitor multiple similar machines (a fleet) operating under comparable conditions.
2. Detect deviations in machine behavior that may indicate potential faults, assuming that most machines are healthy.
3. Operate in an unsupervised manner, eliminating the dependency on pre-labeled datasets.
4. Provide interpretable results and visualizations that allow domain experts to validate and understand anomaly predictions.

Machines with anomaly scores above a predefined threshold are flagged as potentially faulty. The proposed framework addresses this problem through four interconnected modules: **machine comparison**, **fleet clustering**, **anomaly detection**, and **visualization**, enabling a scalable, interpretable, and data-efficient solution for real-time fleet condition monitoring.

5 Proposed Architecture

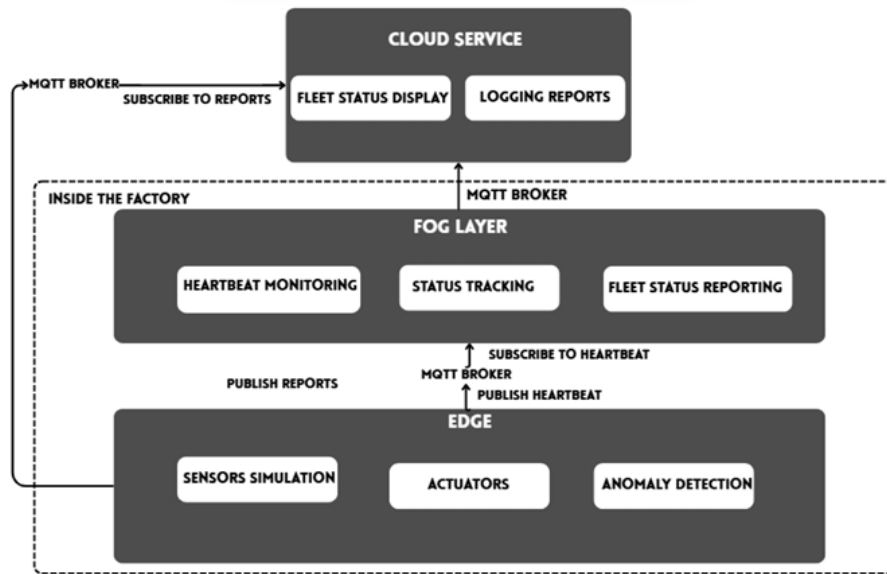


Figure 1: Overview of the Proposed Architecture

Step 1: Machine Comparison

Step 2: Fleet Clustering

Step 3: Anomaly Detection

Step 4: Visualization

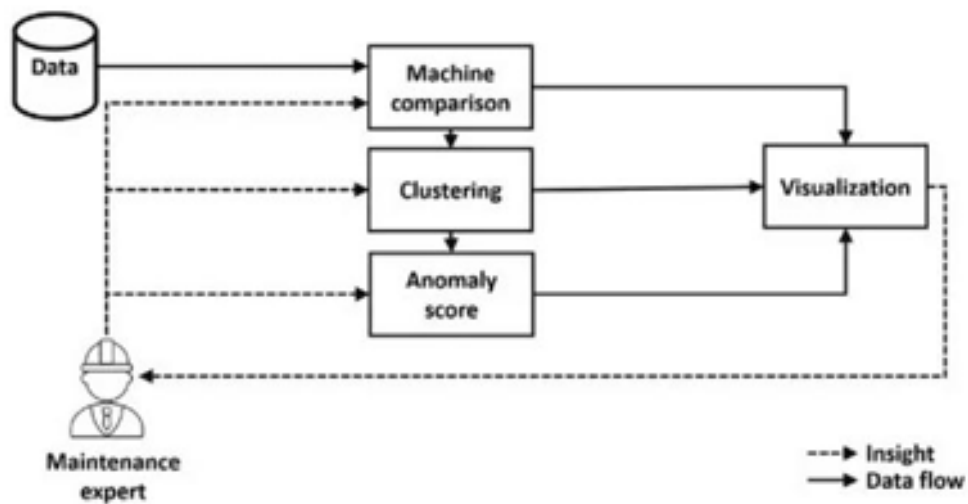


Figure 2: Layered System Design (Edge, Fog, and Cloud Layers)

5.1 Edge Layer

- **Sensor Simulation:** Generates realistic, time-varying data for multiple metrics like temperature, humidity, and pressure.
- **Local Data Buffering:** Maintains a rolling window (local buffer) of recent sensor readings.
- **Edge-based Anomaly Detection:** Performs real-time anomaly detection locally, reducing the need to send all raw data to the cloud.
- **MQTT Communication:** Uses the Message Queuing Telemetry Transport (MQTT) protocol for machine-to-machine (M2M) communication.
- **Publish Reports:** Sends detailed analysis reports (decision, anomaly score, metrics) to the cloud via fog for further processing.
- **Publish Heartbeats:** Sends regular “alive” status messages to the fog layer.
- **Actuator Triggering:** Can trigger actuators autonomously without instruction from cloud or fog.

5.2 Fog Layer

A heartbeat is a small, periodic signal that an edge node sends to a central controller (or monitoring system) to indicate that it is still active and functioning.

- **Heartbeat Monitoring:** Subscribes to a wildcard MQTT topic (edge/+ /heartbeat) to listen for “alive” signals from all edge nodes. The payload might simply be the string “alive”.
- **Status Tracking:** Maintains the status of each edge node. If a node fails to send a heartbeat within a configurable `HEARTBEAT_TIMEOUT`, it is marked as “unresponsive.” When a heartbeat resumes, it is marked as “alive” again.
- **Fleet Status Reporting:** Periodically compiles a “fleet snapshot” summarizing the status of every edge node and publishes it to the cloud.

This mechanism ensures that edge devices remain functional and faults are detected without delay.

5.3 Cloud Layer

- **MQTT Subscriber:** Connects to an MQTT broker and subscribes to:
 - **Reports:** To receive anomaly analysis reports from each fog node.
 - **Fleet_Snapshot:** To receive periodic health summaries for the entire fleet.
- **Responsive UI:** The dashboard automatically refreshes once new data is received, providing a live system view at periodic intervals.
- **Fleet Snapshot:** Displays the overall fleet size and status, including a table and bar chart visualizing anomaly scores for all active nodes.
- **Live Report Stream:** Shows a continuously updating table of detailed reports as they arrive from edge nodes.

6 Methodology

Unsupervised Learning for Anomaly Detection

- **Hierarchical Clustering:** The framework applies agglomerative hierarchical clustering to group machines by pairwise similarity. Each machine begins as its own cluster, and merges occur iteratively based on linkage criteria until a full dendrogram is formed. A **correlation threshold** is then used to cut the dendrogram into meaningful clusters, avoiding arbitrary distance cutoffs and ensuring structure is guided by the data itself.
- **Anomaly Identification:** The assumption is that most machines in the fleet operate under healthy conditions. Hence, large clusters representing the majority of machines are considered *normal*, while smaller clusters — containing fewer machines — are flagged as *anomalous*. This reflects the principle that faults are rare deviations from dominant behavior.
- **Anomaly Scoring:** Each machine is assigned a fraction-based anomaly score that quantifies how isolated it is from the rest of the fleet:

$$\text{Anomaly Score}_i = 1 - \frac{\text{Size of Cluster}_i}{\text{Total Number of Machines}}$$

Scores near zero indicate strong normality (membership in large clusters), while higher scores suggest isolation or abnormal patterns. A threshold — typically around $\frac{2}{3}$ — is used to classify machines as normal or anomalous:

$$\text{Machine}_i = \begin{cases} \text{Normal,} & \text{if Anomaly Score}_i < \frac{2}{3}, \\ \text{Anomalous,} & \text{otherwise.} \end{cases}$$

- **Visualization and Interpretability:** To ensure transparency and explainability, several plots are generated to visualize clustering and anomalies:
 - **Normalized Signal Plots:** Compare individual machine signals on a common scale.
 - **Dissimilarity Matrix:** Displays pairwise distances and highlights clusters and outliers.
 - **Dendrogram:** Shows hierarchical relationships and the correlation cut threshold.
 - **Anomaly Score Distribution:** Provides a fleet-level overview of machine health.

- **Benchmark Method:** For performance evaluation, the framework is compared against a simpler statistical rule-based approach. In the benchmark, a machine is marked anomalous if its measurement deviates by more than a few standard deviations from the fleet mean:

$$|x_i - \mu| > k\sigma, \quad \text{where } k \text{ is a sensitivity parameter.}$$

This comparison highlights the improved adaptability and interpretability of the clustering-based method over conventional threshold-based rules.

- **Summary:** The hierarchical clustering framework provides a principled, transparent, and data-driven approach to anomaly detection. Its emphasis on visualization and interpretability makes it well-suited for industrial monitoring and predictive maintenance applications.

7 Results and Discussions

7.1 Accuracy

The performance of the proposed unsupervised anomaly detection model shows a significant improvement as the fleet size increases. When evaluated on a smaller fleet of 50 machines, the model achieved an accuracy of 93%. However, as the fleet size expanded to 300 machines, the accuracy rose to 97%. This improvement can be attributed to the greater diversity and volume of sensor data available in larger fleets, which allows the clustering and similarity-based algorithms to better learn the representation of normal machine behavior. The increased data variety enhances the model's ability to distinguish between true anomalies and natural operational variations, leading to more stable and reliable fault detection across the system.

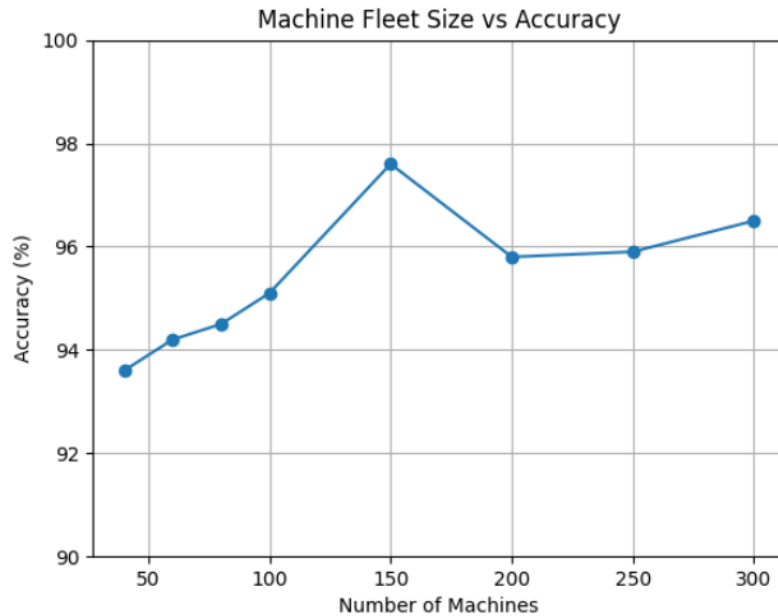


Figure 3: Machine fleet size vs. accuracy.

7.2 End-to-End Latency

End-to-End Latency: The system's transmission performance was evaluated based on message size and network capacity.

- **Message size for 100 machines:** 180 KB
- **Network transmission rate:** 100 Mbps

The propagation delay is considered negligible compared to the transmission delay.

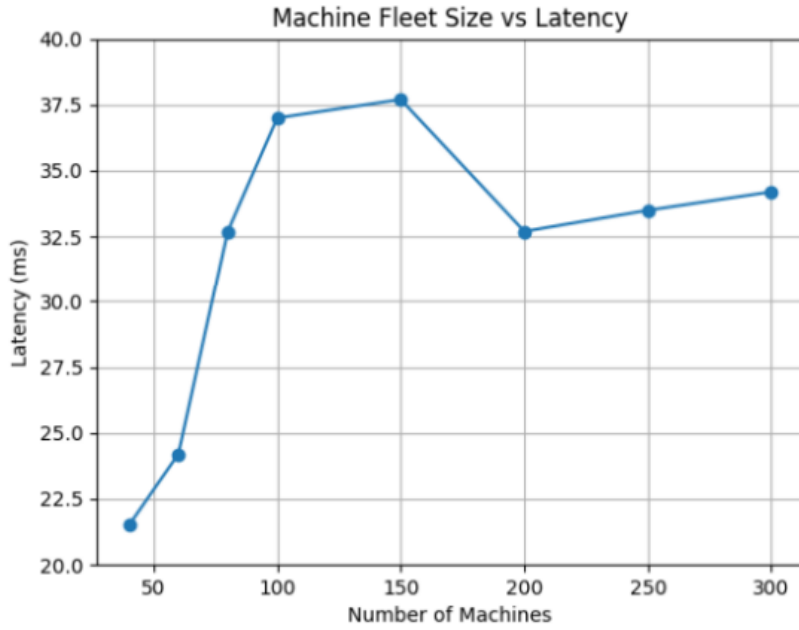


Figure 4: Fleet size vs. end-to-end latency.

7.3 Processing power

When the number of machines grew from 50 to 300, the CPU utilization of the edge node rose from 10% to 40%. This rise is expected, as a larger fleet generates more sensor data that must be processed locally for clustering and anomaly detection. The utilization remains within acceptable limits for real-time edge operations, demonstrating that the proposed framework maintains reliable performance.

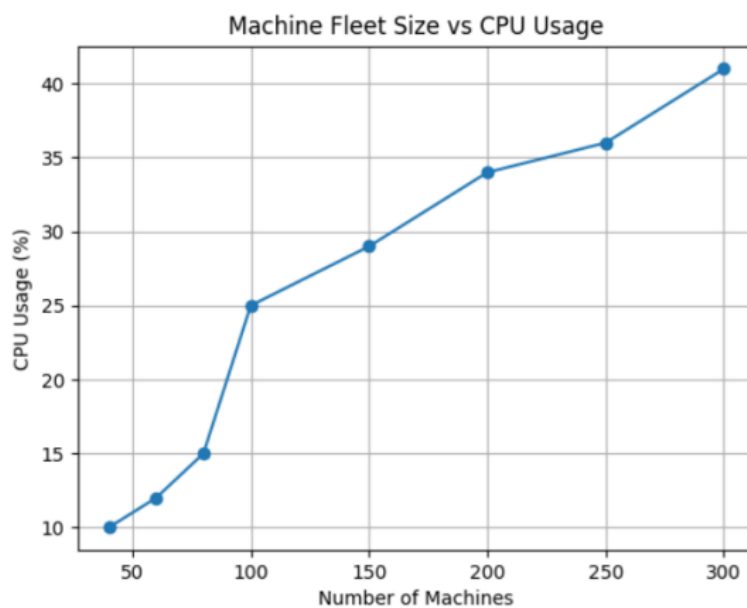


Figure 5: Fleet size vs. CPU utilization.

The RAM utilization followed a similar upward trend, increasing from 10 MB to 140 MB as the fleet size expanded from 50 to 300 machines. A sudden spike in memory usage was observed around 200 machines, which corresponds to the point where the system begins handling multiple large data buffers simultaneously. This behavior can be attributed to the higher data throughput and the growing number of active clustering computations required for real-time anomaly detection.

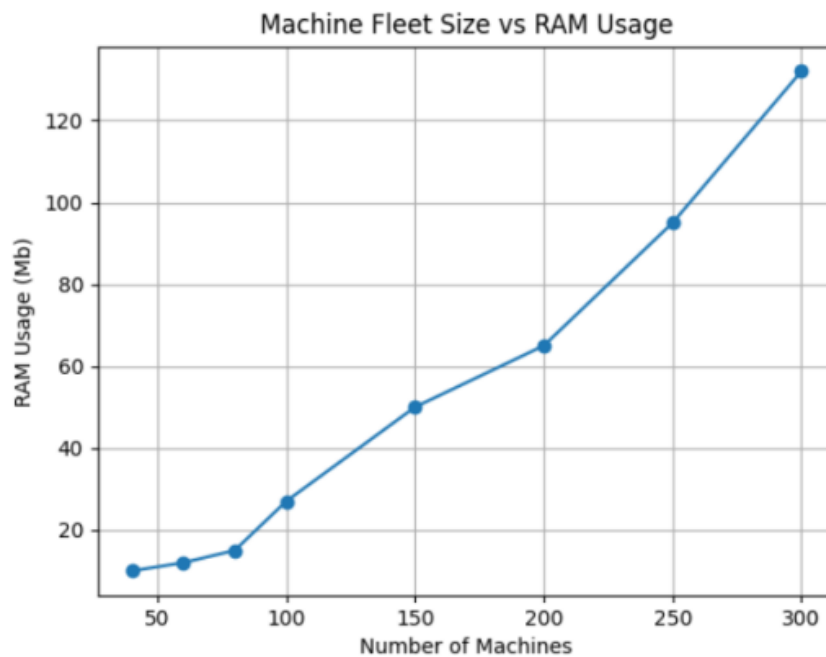


Figure 6: Fleet size vs. RAM utilization.

7.4 Fleet Snapshot

What is shown: A compact table listing each edge node, the time of the latest message, and the machine indices flagged as anomalous by local models.

Why it matters: Operators can immediately identify which nodes require attention and start investigation workflows. It is the first level of triage that prioritizes limited maintenance resources.

Operational insight: This view confirms that the simulated edges publish regularly and that the central UI collects and displays metadata stores reliably. It also demonstrates the value of showing flagged indices rather than raw telemetry for quick triage.

Fleet Snapshot

Timestamp: 2025-10-20T11:26:17.712916Z | Edge nodes: 3

Node Status:

	edge_id	status	n_machines
0	edge3	alive	100
1	edge1	alive	100
2	edge2	alive	100

Anomalous Machines per Edge Node:

	edge_id	anomalous_machines
edge3	edge3	85
edge1	edge1	3 7 13 14 20 30 41 48 56 74 95
edge2	edge2	8 32 35 48 94 97

Figure 7: Fleet Snapshot: per-edge node status, timestamps and anomalous machine indices.

7.5 Live report stream

What is shown: The raw stream of metric vectors (normalized), each associated with a timestamp and an edge identifier.

Why it matters: During development and debugging this stream verifies message order, latency, and parsing. For operations, it can indicate transient anomalies correlated to specific events like startup spikes.

Operational insight: A stable stream with little jitter indicates the messaging and edge simulators are performing reliably. Missing rows or large gaps would point to networking issues or crashed publishers.

Live Report Stream

	edge_id	ts	metrics
0	edge2	2025-10-20T11:27:33.778953Z	-0.0348894450989425 0.0983703433351799 0.058092283071714 0.007028444
1	edge1	2025-10-20T11:27:33.669155Z	0.0496714153011232 -0.0138264301171184 0.0647688538100692 0.53152257
2	edge3	2025-10-20T11:27:33.659665Z	-0.0091949919865191 -0.1463350652811679 0.1081791679198325 -0.0239325
3	edge2	2025-10-20T11:27:28.769146Z	-0.0348894450989425 0.0983703433351799 0.058092283071714 0.007028444
4	edge1	2025-10-20T11:27:28.665529Z	0.0496714153011232 -0.0138264301171184 0.0647688538100692 0.53152257
5	edge3	2025-10-20T11:27:28.654783Z	-0.0091949919865191 -0.1463350652811679 0.1081791679198325 -0.0239325
6	edge2	2025-10-20T11:27:23.758775Z	-0.0348894450989425 0.0983703433351799 0.058092283071714 0.007028444
7	edge1	2025-10-20T11:27:23.662076Z	0.0496714153011232 -0.0138264301171184 0.0647688538100692 0.53152257
8	edge3	2025-10-20T11:27:23.649575Z	-0.0091949919865191 -0.1463350652811679 0.1081791679198325 -0.0239325
9	edge2	2025-10-20T11:27:18.747653Z	-0.0348894450989425 0.0983703433351799 0.058092283071714 0.007028444
10	edge1	2025-10-20T11:27:18.658648Z	0.0496714153011232 -0.0138264301171184 0.0647688538100692 0.53152257
11	edge3	2025-10-20T11:27:18.638299Z	-0.0091949919865191 -0.1463350652811679 0.1081791679198325 -0.0239325
12	edge2	2025-10-20T11:27:13.742059Z	-0.0348894450989425 0.0983703433351799 0.058092283071714 0.007028444

Figure 8: Live Report Stream: timestamped metric vectors from edge nodes.

8 Conclusion

This research presents an edge-based anomaly detection framework for fleet-based condition monitoring of machines. The proposed edge-fog-cloud architecture enables real-time fault detection at the network edge while reducing latency, bandwidth consumption, and data privacy risks. By leveraging unsupervised anomaly detection through hierarchical clustering, the framework successfully identifies machine anomalies without requiring historical labeled failure data. Experimental simulations demonstrate that the distributed architecture achieves low-latency fault detection, scalable fleet management, and effective anomaly visualization suitable for industrial IoT environments with limited connectivity.

9 Future Work

- Integration of deep learning or hybrid models to improve anomaly detection accuracy.
- Optimization of edge node resources (CPU, RAM, energy) for handling larger fleets efficiently.
- Real-world deployment and testing on actual industrial machines to validate performance.
- Incorporation of security measures, such as encryption and authentication, for safe data communication.

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